Basic Programming Algorithms – SIMPLE TASKS

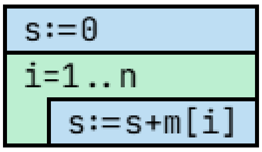
Specification tool: *https://progalap.elte.hu/specifikacio/*)
Algorithm tool: *https://progalap.elte.hu/stuki/*)

1. Summation

Specification for simpler task

In: n∈Z, m∈Z[1..n]
Out: s∈Z
Pre: -
Post: s=SUM(i=1..n, m[i])

Algorithm for simpler task

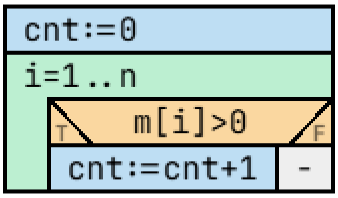


2. Counting

Specification for simpler task

(Count all the positive numbers (m[i]>0])
In: n∈Z, m∈Z[1..n]
Out: cnt∈N
Pre: -
Post: cnt=COUNT(i=1..n, m[i]>0)

Algorithm for simpler task

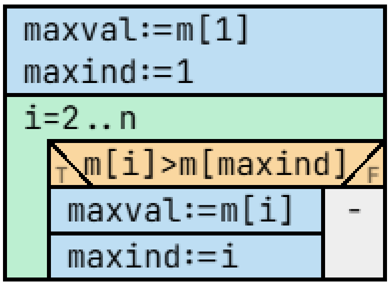


3. Maximum selection

Specification for simpler task

(Pick the tallest people)
In: n∈Z, m∈Z[1..n]
Out: maxind∈Z, maxval∈Z
Pre: n > 0
Post:(maxind, maxval) =
MAX(i=1..n, m[i])

Algorithm for simpler task

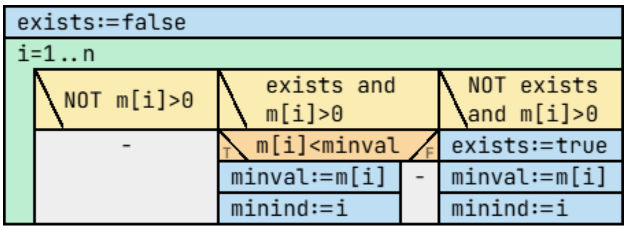


4. Minimum selection with condition

Specification for simpler task

(Pick the smallest from positiv values)
In: n∈Z, m∈Z[1..n]
Out: maxind∈Z, minval∈Z
Pre: n > 0
Post:(exists, minind, minval) =
MINIF(i=1..n, m[i], m[i]>0)

Algorithm for simpler task

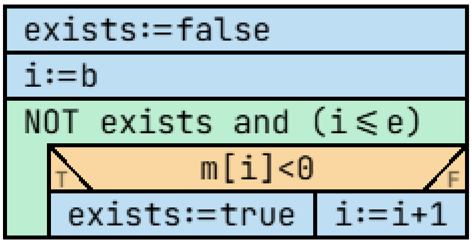


5. Decision

Specification for simpler task

(Determine if array has negative element)
In: n∈Z, m∈Z[1..n]
Out: exists∈L
Pre: -
Post: exists=**EXISTS**(i=1..n, m[i]<0)

Algorithm for simpler task

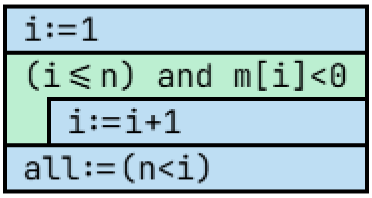


6. Decision all

Specification for simpler task

(Determine all elements are negative)
In: n∈Z, m∈Z[1..n]
Out: all∈L
Pre: -
Post: all=**FORALL**(i=1..n, m[i]<0)

Algorithm for simpler task

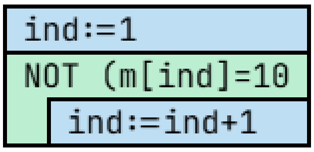


7. Selection

Specification for simpler task

(Select the first element's index equals to 10)
***In:** n∈N, m∈N[1..n]*
***Out:** ind∈N*
***Pre:** n ≥ 1 and ∃i∈[1..n]:(m[i]=10)*
***Post:** ind = **SELECT**(i>=1, m[i] = 10)*
and ind≤n

Algorithm for simpler task

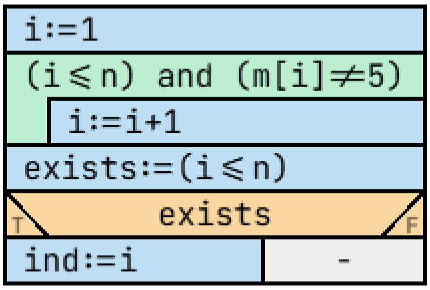


8. Search

Specification for simpler task

(Some numbers in array, search for the number 5. We do NOT know it is exist or not in an array.)
In: n∈Z, m∈Z[1..n]
Out: exists∈L, ind∈Z
Pre: -
Post: (exists, ind)=**SEARCH**(i=1..n,
m[i]=5)

Algorithm for simpler task

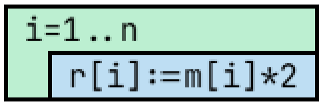


9. Copy

Specification for simpler task

(Store doubled value)
In: n∈Z, m∈Z[1..n]
Out: r∈Z[1..n]
Pre: -
Post: r=COPY(i=1..n, m[i]*2)

Algorithm for simpler task



10. Multiple item selection

Specification for simpler task

(Store doubled value)
In: n ∈ Z, m∈Z[1..n]
Out: cnt∈N, y∈Z[1..cnt]
Pre: -
Post: (cnt, y)=MIS(i=1..n,(m[i]%2=0),
m[i])

Algorithm for simpler task